Week 1 : Assignment 1

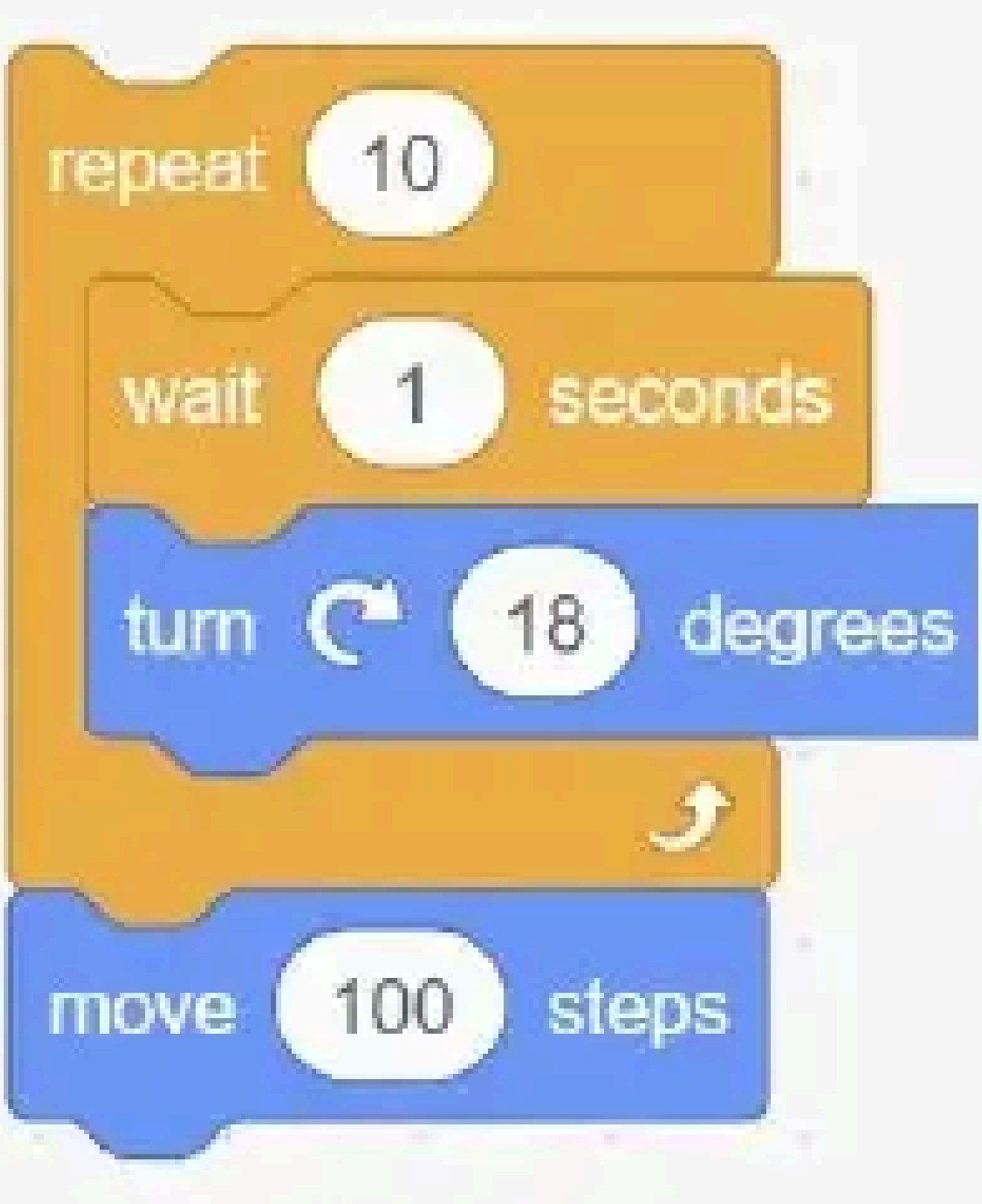
Assignment not submitted

Due date: 2024-08-07, 23:59 IST.

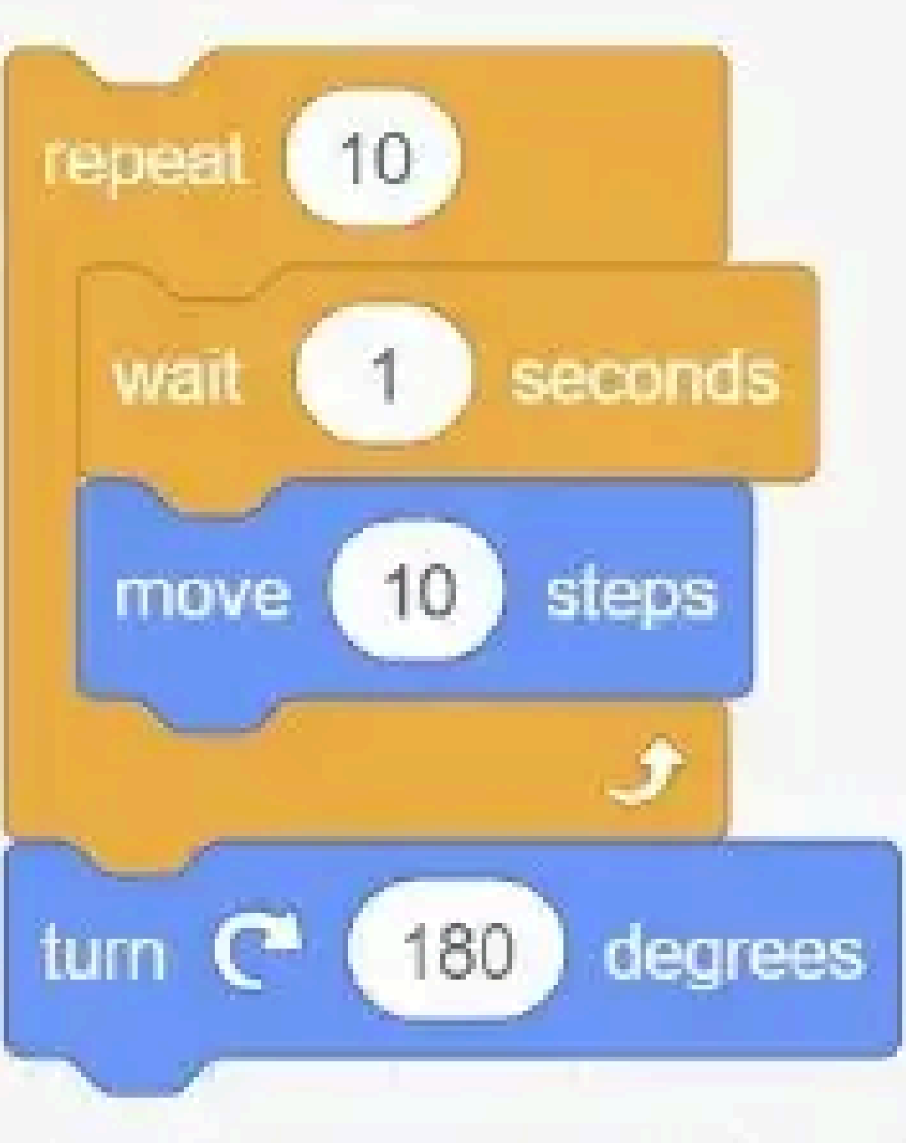
1) The cat wants to move 100 steps away slowly, and by the time it finishes moving 100 steps, it needs to slowly rotate as well towards the opposite direction. Can both these tasks be accomplished by using single loop in scratch? ***1 point***

○ Yes
○ No

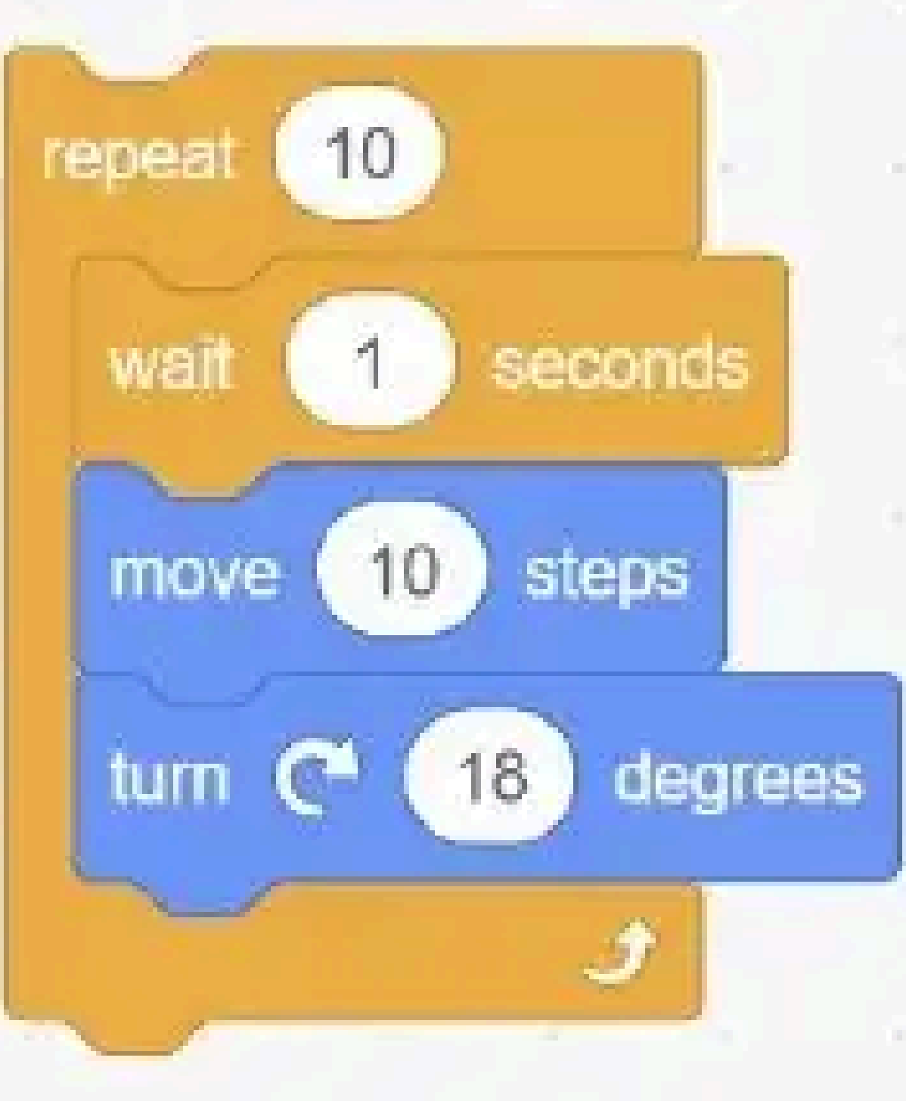
2) Which block moves the cat as described in the previous question ? ***1 point***

○


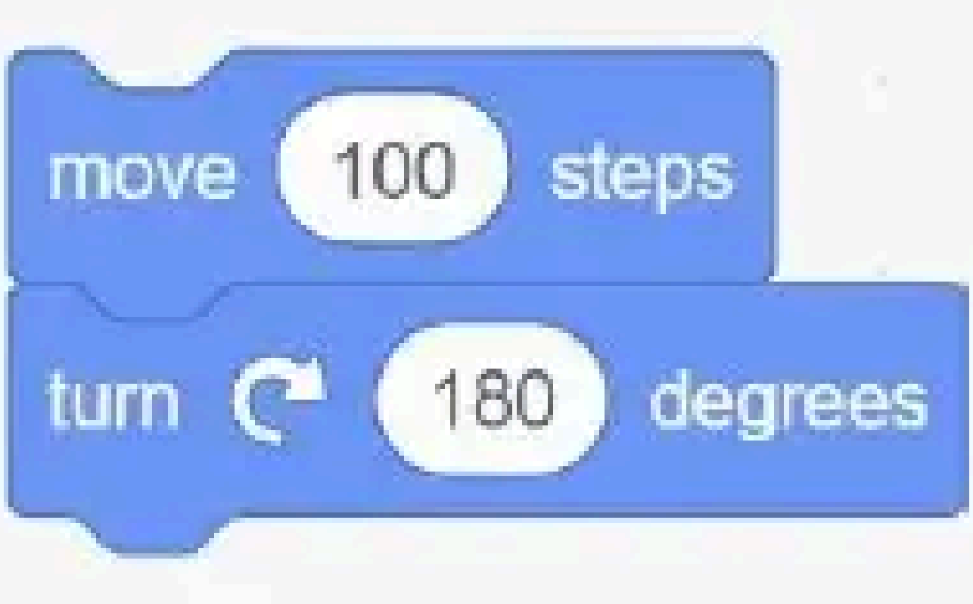
```
repeat 10
    wait 1 seconds
    turn ↻ 18 degrees
move 100 steps
```

○


```
repeat 10
    wait 1 seconds
    move 10 steps
turn ↻ 180 degrees
```

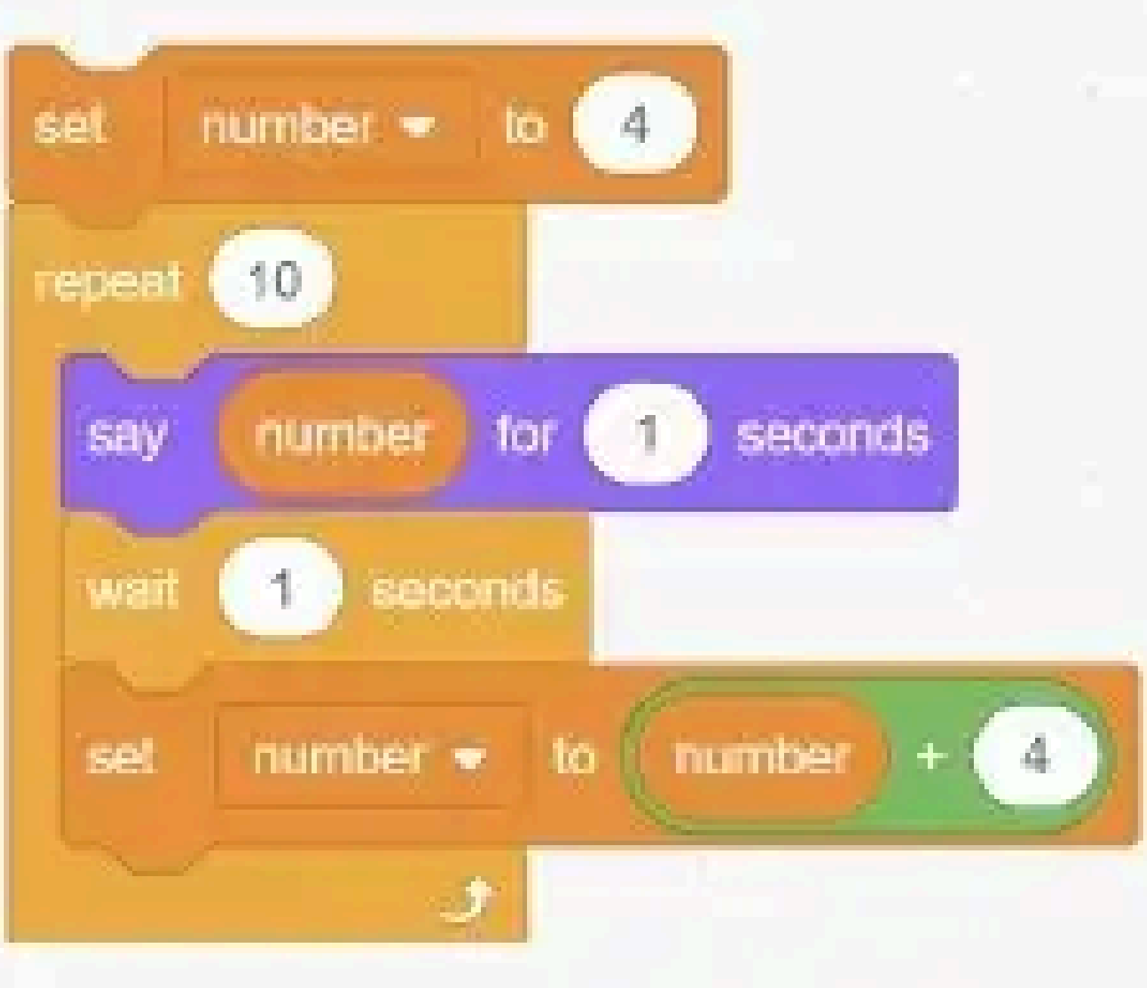
○


```
repeat 10
    wait 1 seconds
    move 10 steps
    turn ↻ 18 degrees
```

○


```
move 100 steps
turn ↻ 180 degrees
```

3) What does the cat say here ? ***1 point***



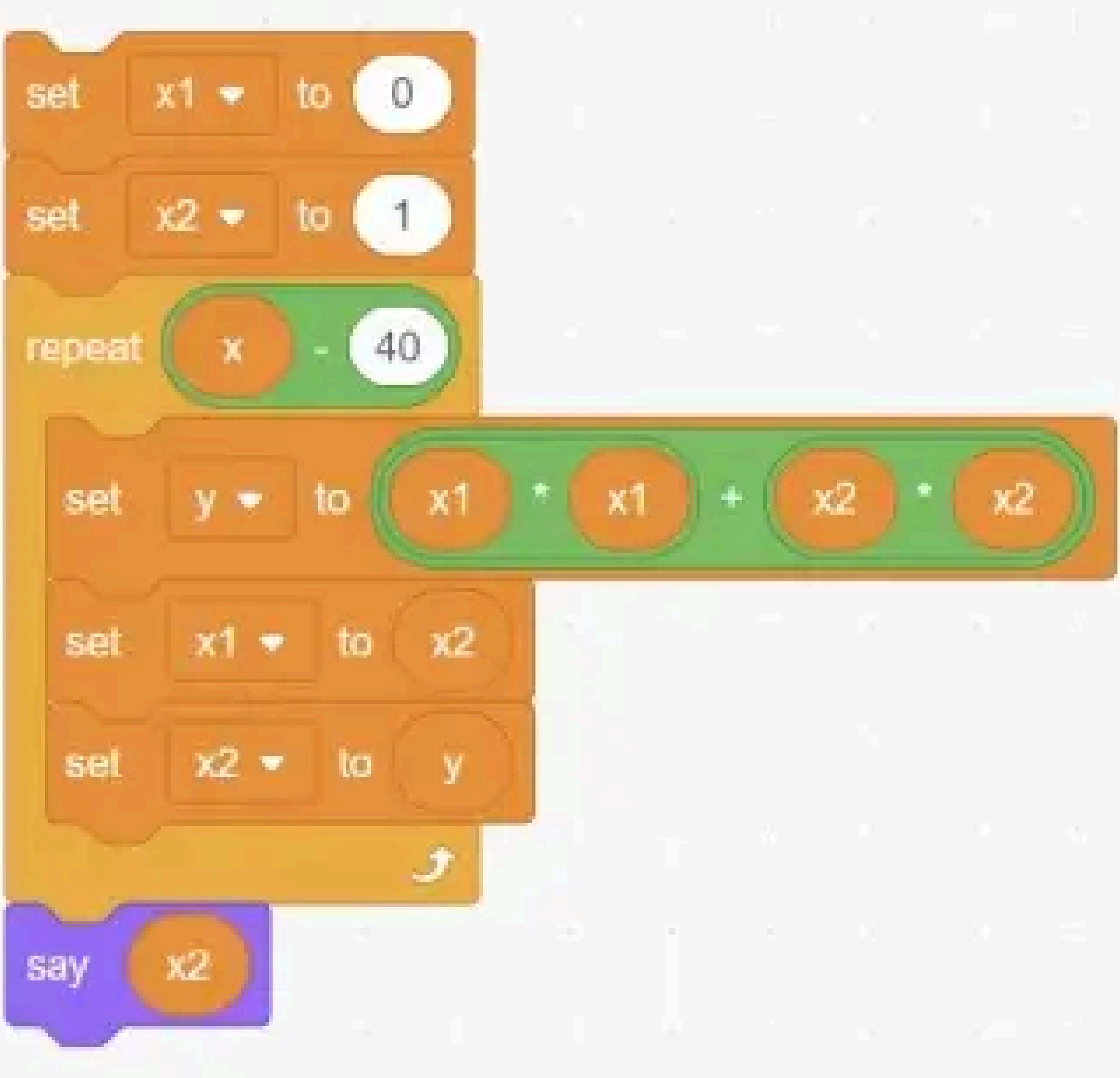
```
set number ▾ to 4
repeat 10
    say number for 1 seconds
    wait 1 seconds
    set number ▾ to number + 4
```

○ Says the number 4, for 1 second, 10 times and changes the value of number variable by multiplying previous value with 4.
○ Says the number 4, for 1 second, 10 times and changes the value of number variable by adding previous value with 4.
○ Says the number 4, and its multiples each for 1 second, till 40 and changes the value of number variable by multiplying previous value with 4.
○ Says the number 4, and its multiples each for 1 second, till 40 and changes the value of number variable by adding previous value with 4.

4) From the previous question, what is the value of number variable after the loop ends ? ***1 point***

○ 40
○ 44
○ 36
○ 4

5) Let answer for previous question be stored in variable x, what does the cat say after the execution of the loop ? ***1 point***



```
set x1 ▾ to 0
set x2 ▾ to 1
repeat x - 40
    set y ▾ to x1 * x1 + x2 * x2
    set x1 ▾ to x2
    set x2 ▾ to y
say x2
```

○ 16
○ 5
○ 29
○ 44

6) Which of the following is use case of variables in programming. ***1 point***

○ They are used to make the computer store program data into SSD.
○ They help in storing and retrieval of information/data while a task on the machine is running.
○ They simplify a task which involves processing repetitive procedures.They simplify a task which involves processing repetitive procedures.
○ They help in running multiple tasks parallel to each other.

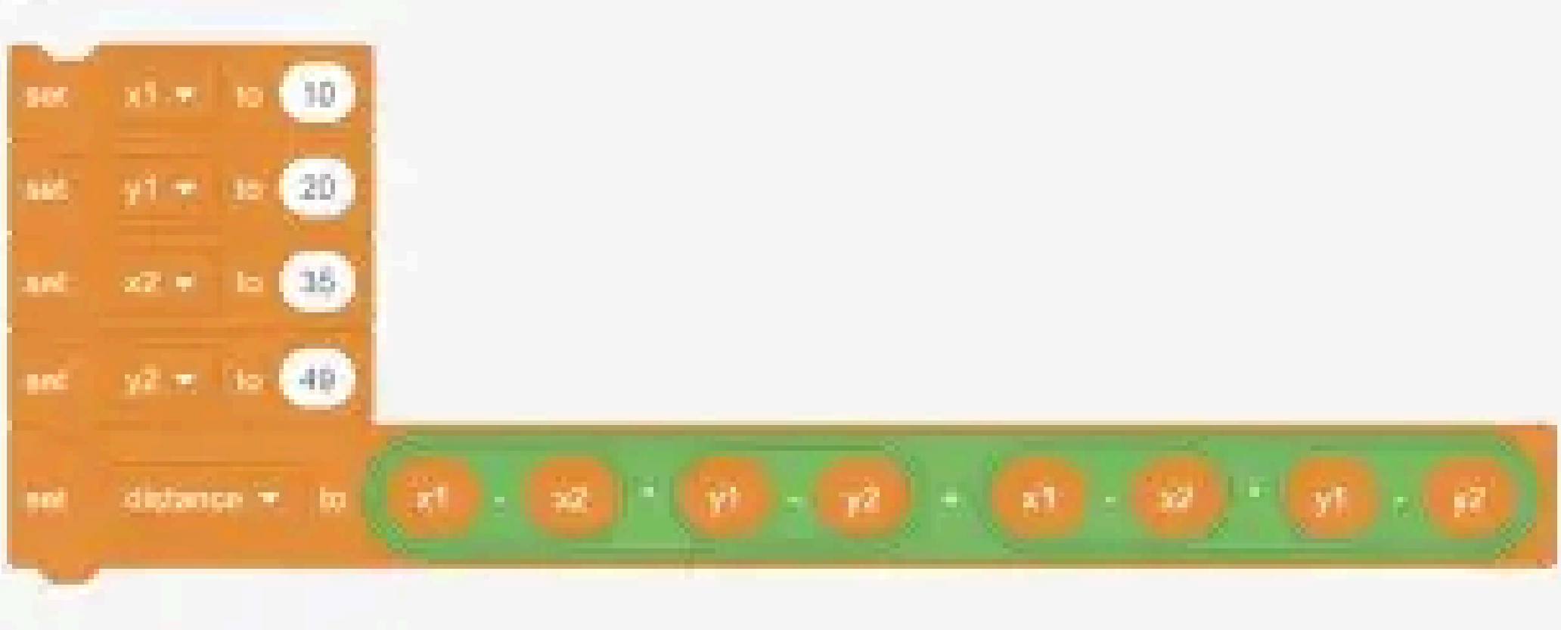
7) What is the concept that helps in instructing the computer to execute repetitive tasks ? ***1 point***

○ Functions
○ Variables
○ Conditionals
○ Loops

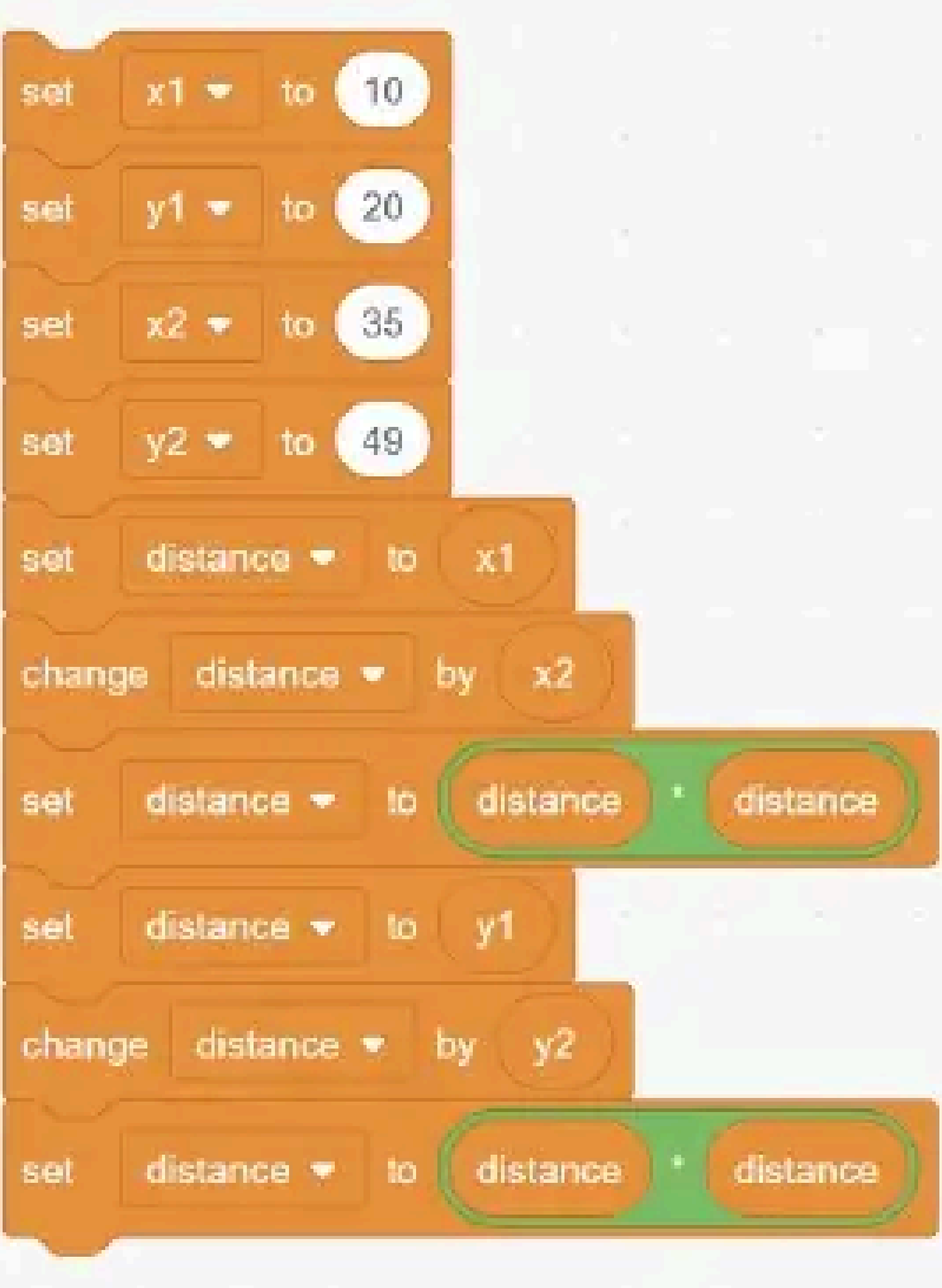
8) Can polynomial expressions be instructed to computer in logically finite steps ? ***1 point***

○ Yes
○ No

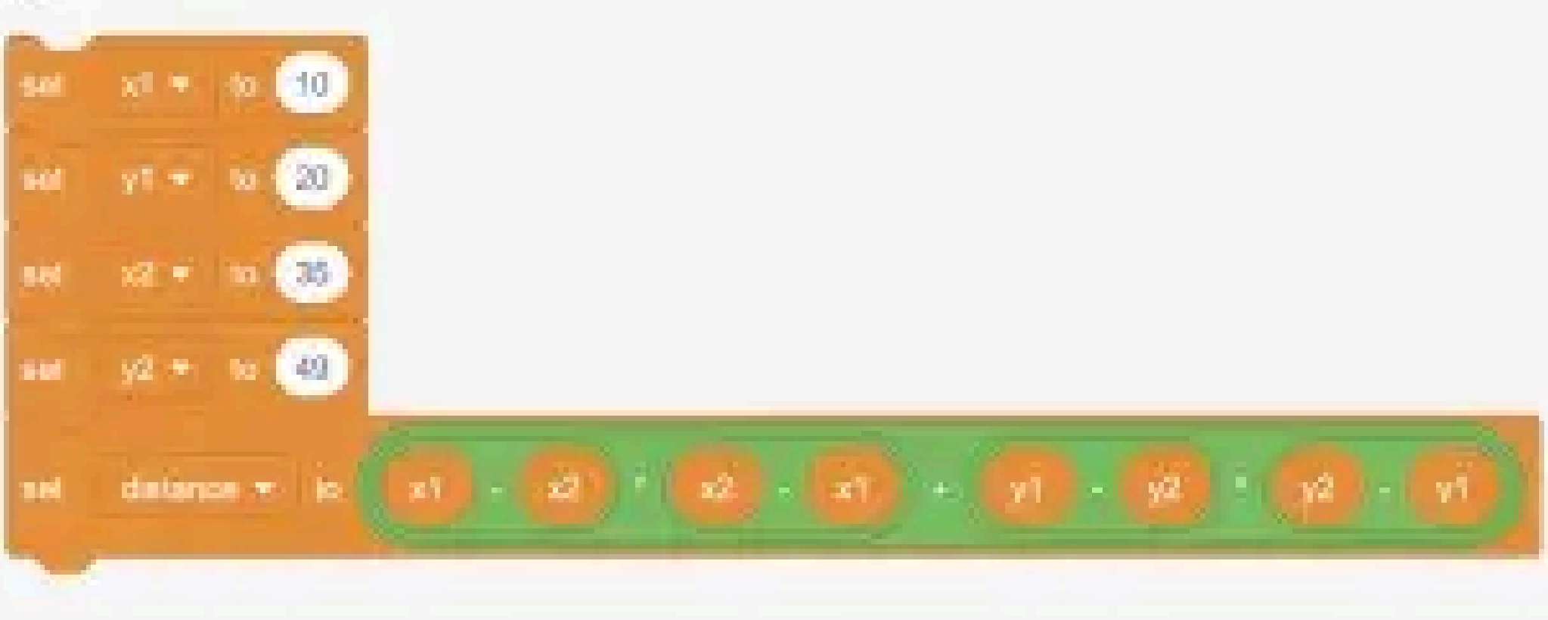
9) In which of the following code blocks **calculation** of the squared distance between two points (10, 20) and (35, 49) is accomplished. ***1 point***

○


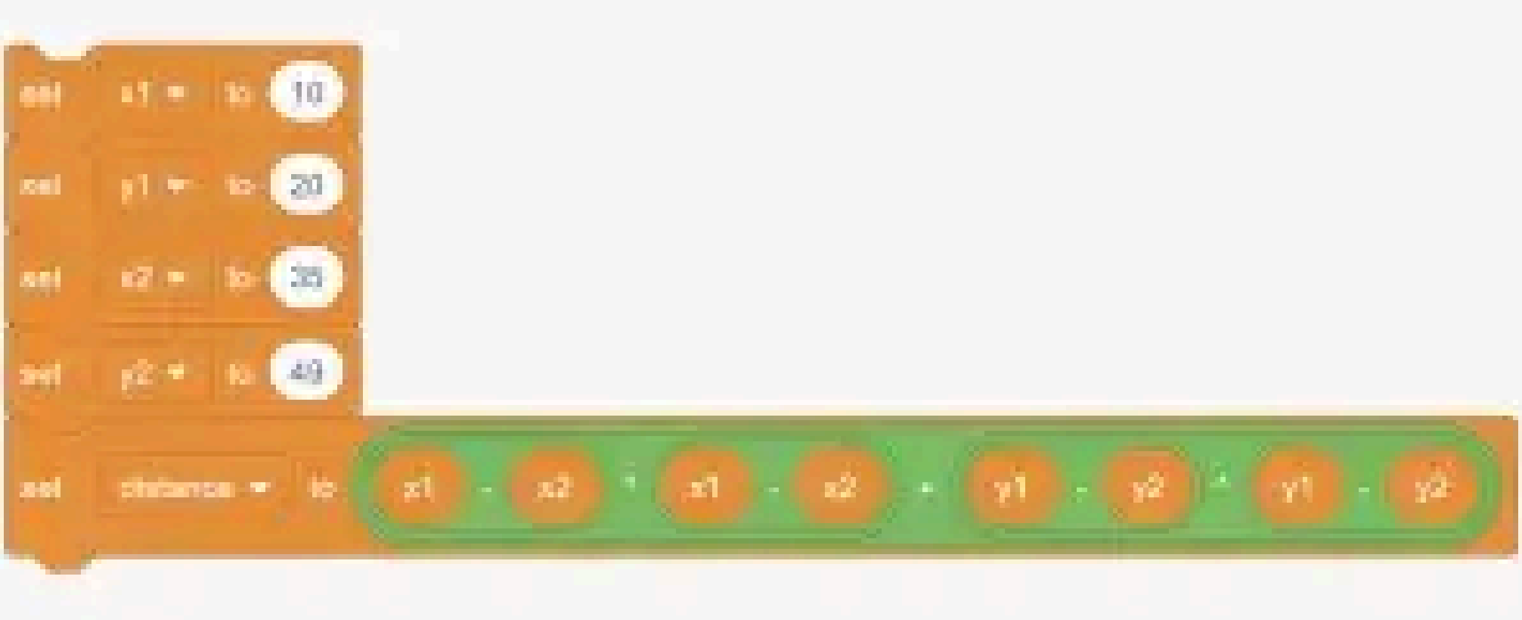
```
set x1 ▾ to 10
set y1 ▾ to 20
set x2 ▾ to 35
set y2 ▾ to 49
set distance ▾ to x1 - x2 * y1 - y2 + x1 - x2 * y1 - y2
```

○


```
set x1 ▾ to 10
set y1 ▾ to 20
set x2 ▾ to 35
set y2 ▾ to 49
set distance ▾ to x1
change distance ▾ by x2
set distance ▾ to distance * distance
set distance ▾ to y1
change distance ▾ by y2
set distance ▾ to distance * distance
```

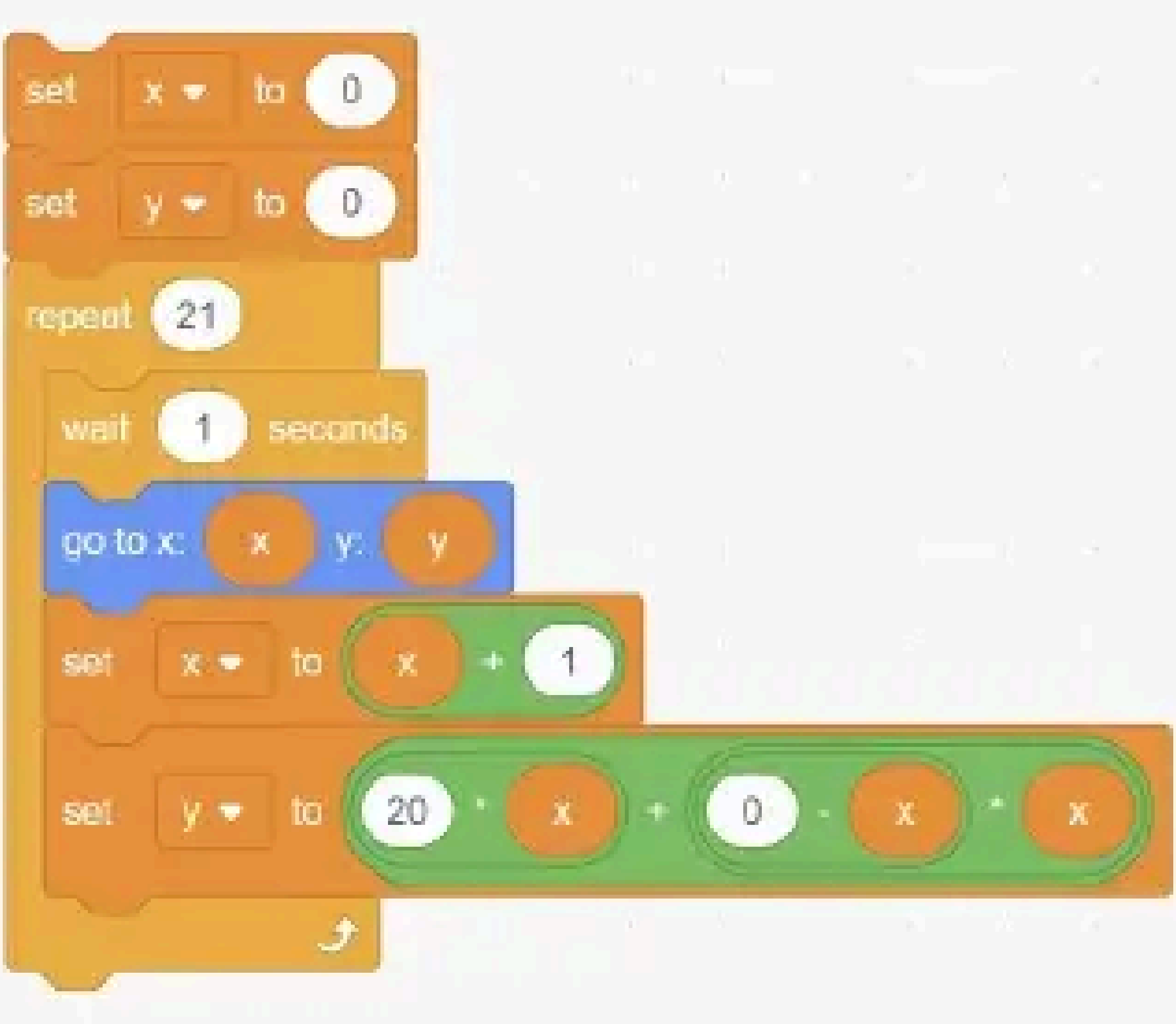
○


```
set x1 ▾ to 10
set y1 ▾ to 20
set x2 ▾ to 35
set y2 ▾ to 49
set distance ▾ to x1 - x2 * x2 - x1 + y1 - y2 * y2 - y1
```

○


```
set x1 ▾ to 10
set y1 ▾ to 20
set x2 ▾ to 35
set y2 ▾ to 49
set distance ▾ to x1 - x2 * x1 - x2 + y1 - y2 * y1 - y2
```

10)What task does the block below accomplish ? ***1 point***



```
set x ▾ to 0
set y ▾ to 0
repeat 21
    wait 1 seconds
    go to x: x y: y
    set x ▾ to x + 1
    set y ▾ to 20 * x + 0 - x * x
```

○ Moves the ball in a parabolic path and stops 21 away steps from center(0, 0) in x direction.
○ Moves the ball in a circular path and 21 times.
○ Moves the ball in a triangular path and end at the center (0, 0).
○ Moves the ball in a parabolic path and stops 20 away steps from center(0, 0) in x direction.

You may submit any number of times before the due date. The final submission will be considered for grading.

Submit Answers